

Prashant Bhattarai

Starkville, MS | 662 518 1958 | prashantbhattarai224@gmail.com | github.com/prashant-bhattarai

EDUCATION

Mississippi State University

B.S. in Mechanical Engineering — GPA: 4.00/4.00

Starkville, MS

Expected May 2028

- **Relevant Coursework:** Modeling & Manufacturing, Engineering Mechanics, Circuits, Fluid Mechanics, Thermodynamics

EXPERIENCE

EcoCAR Innovation Challenge Program, Mississippi State University

Co-Lead, Vehicle Hardware Integration

Starkville, MS

August 2026 – Present

- Engineer EcoCAR vehicle system integrations in **Siemens NX** and deploy a custom **GitLab** version control workflow to manage shared CAD assemblies across the university's HPC environment.

Center for Advanced Vehicular Systems (CAVS), Mississippi State University

Undergraduate Researcher

Starkville, MS

January 2026 – Present

- Lead UAV-assisted map recovery research for autonomous ground vehicles, achieving an 80% success rate across 50 **MAVS** simulation trials before deploying hardware on a **Jetson**-powered RC car platform using **Nav2** and **DLIO** LiDAR-IMU SLAM.
- Engineer a perception-based leader-follower architecture for off-road convoys with minimal V2V communication, utilizing **Camera-LiDAR fusion**, advancing from ArUco to ML vehicle detection, while maintaining independent local autonomy.
- Design mechanical sensors mounts and server heat exhausts for military-grade autonomous vehicles in **SolidWorks**, executing full embedded hardware integration (**Jetson**, LiDAR, camera, IMU) for 1/8-scale RC platforms applying **DFM** principles.
- Benchmarked centralized vs. decentralized multi-agent navigation using CBS path planning, MAVS, and the NATURE stack.

USDA REEU – AI2F Summer Research Program, Mississippi State University

Product Design Research Intern

Starkville, MS

June 2025 – August 2025

- Engineered a modular, AI-enabled field device from concept to deployment using **SolidWorks**, delivering a **fully integrated hardware prototype** for downstream PhD research and presenting findings at an undergraduate symposium.

Industrial & Systems Engineering, Mississippi State University

Undergraduate Research Assistant

Starkville, MS

August 2025 – November 2025

- Designed **structural lattice components** using FDM, resin, and **metal additive manufacturing** to directly support BCM research on improving ceramic nanocomposite fracture toughness, concurrently managing lab procurement and technical AM workshops.

ISTVS Student Chapter, Mississippi State University

Undergraduate Research Assistant

Starkville, MS

February 2025 – April 2025

- Benchmarked GPS-only vs. **GPS/IMU sensor fusion** localization on a Polaris MRZR in off-road environments.

PROJECTS

• Computer Vision & Robotics Perception | C++, OpenCV

Developed a C++ perception pipeline for AMR docking using **OpenCV's solvePnP** on multi-marker ArUco grids for real-time **6-DoF pose estimation** under heavy occlusion, fused with a 6-state **Kalman filter** to eliminate tracking jitter. Built a vision-based safety interlock using MOG2 background subtraction and centroid tracking, cutting processed pixel area over 60%.

• Synthetic Data Generation & Object Detection | NVIDIA Isaac Sim, YOLO11, Python, ROS2

Ongoing

Built a synthetic-data pipeline in **NVIDIA Isaac Sim Replicator** and trained a **YOLO11s** detector for fallen boxes (0.99 val mAP50-95). Diagnosed a sim-to-real gap and am iterating on the dataset while building a **ROS2** inference node.

• Non-Linear Model Predictive Control (MPC) | Python, NumPy, CasADi

Ongoing

Derived non-linear state-space dynamics for an inverted pendulum from first-principles mechanics and built a custom **RK4** physics simulation engine. Formulated a discrete-time non-linear **MPC** controller in **CasADi** solving constrained real-time optimization, enforcing state/input limits for hardware deployment.

LEADERSHIP

• Chief of Additive Manufacturing, Advanced Manufacturing Society

Feb 2025–Present

Led hands-on technical workshops focused on applied engineering skills, additive manufacturing, 3D modeling, and manual/CNC machining, exposing 250+ students to advanced industry fabrication opportunities and SolidWorks certification training.

• Vice President, International Society for Terrain-Vehicle Systems (ISTVS)

Apr 2025–Present

Co-organizing the first ISTVS competition with Virginia Tech and Pretoria, concurrently modeling wheel designs in SolidWorks and executing YADE DEM simulations to analyze rolling resistance and optimal angular velocity for maximum drawbar pull.

SKILLS

Robotics & Autonomy: ROS2, NATURE, NAV2, MAVS, Sensor Fusion, Hardware Integration, Path Planning, SLAM

Programming & Software: Python, C/C++, Linux, Docker/Podman, Git, OpenCV, NumPy, CasADi, Embedded Systems

Mechanical & Manufacturing: SolidWorks (CSWP), Siemens NX, YADE DEM, FEA, GD&T, DFM, CNC, Rapid Prototyping